VISHWAKARMA INSTITUTES VI

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

CODE APEX

HACK THE UPSIDE DOWN

Team Details

- Team name - Code Sprints
- Team leader Name - Shaheen Khan
- Problem Statement - Fake News and Misinformation Detection

Brief about the idea

65%

of people encounter fake news weekly

6x

Faster fake news spreads vs real news

3.6B

Social media users exposed daily

Existing Tools FAIL at:	VS	TruthLens SOLVES this:
✗ Only give a score — no explanation of WHY		✓ Claim-by-claim autopsy with exact sentence highlight
✗ Miss psychological manipulation tactics		✓ Detects fear, urgency & false authority patterns
✗ English-only — no Indian language support		✓ Hindi, Marathi, Gujarati, Telugu, English support
✗ Cannot trace rumour origins or mutations		✓ Misinformation Family Tree across platforms

Proposed Solution Framework

01

Misinformation Autopsy

- Breaks article into verifiable claims.
- Highlights exact false sentence.

02

Manipulation Fingerprint

- Detects fear, urgency & false authority.
- Visualised as a radar chart.

03

Source DNA Check

- Domain age, author history, cross-referenced
- with fact-check databases via WHOIS.

04

ELI5 Explainer

- Converts full autopsy into 2 plain sentences
- any non-technical reader understands.

05

Misinfo Family Tree

- Traces if story is a mutation of an older hoax.
- Shows 2019–2024 spread chain.

Credibility Score:

61–100 CREDIBLE

Technical Approach & System Architecture

INPUT
Text / Image
URL / Screenshot

OCR
Tesseract.js
In-browser

NLP ENGINE
BERT + Gemini
Claim Extraction

VERIFY
Fact Check API
WHOIS + NewsAPI

OUTPUT
Autopsy Card
0–100 Score

NLP Analysis Pipeline

1

Claim Decomposition

Gemini API splits article into atomic verifiable facts

2

Manipulation Scoring

BERT on LIAR + FakeNewsNet detects fear & urgency

3

Source DNA Check

WHOIS domain age + historical fake-news database lookup

4

Origin Tracing

NewsAPI cross-references past hoax mutations

Technology Stack of the Solution

Frontend

- React.js
- Tailwind CSS
- Tesseract.js OCR

AI & NLP

- Gemini 1.5 Flash
- BERT (fine-tuned)
- Google Fact Check API

Source Verify

- WHOIS API
- NewsAPI
- Fake News DB
- 74K Indian Dataset

Deployment

- Vercel (frontend)
- Chrome Extension MV3
- No backend for MVP

Implementation Roadmap

9 hours • 4 phases • 1 product

01 Core Foundation

- React + Tailwind scaffold
- Gemini API integration
- Credibility score UI

02 USP Features

- Manipulation Fingerprint
- WhatsApp OCR upload
- ELI5 explainer button

03 Polish & Deploy

- Autopsy card output UI
- Fact Check API wired
- Vercel deployment

04 Bonus Features

- Misinfo Family Tree
- Hindi auto-detect
- Spread Tracker graph

Expected Impact & Scalability

< 5 sec

Full autopsy per article

100%

Explainable AI — no black box

5 langs

Indian languages supported

Scalability Roadmap

MVP	**Hackathon:**	Web app — paste text or upload WhatsApp screenshot for instant autopsy
V2	**Post-Hack:**	Browser extension — live credibility badge on every news site
V3	**Scale:**	REST API service for newsrooms, fact-checkers & social platforms
VISION	**Long-term:**	Real-time WhatsApp bot for rural India in regional languages

Ethical Commitment

- Flags content — never censors it
- Reader always makes the final call
- Tested across all political viewpoints
- Zero black-box verdicts — all explained

VISHWAKARMA INSTITUTES VI

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

VCA

CODE APEX

HACK THE UPSIDE DOWN

THANK YOU